



The resilience of entrepreneurial ecosystems



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ABSTRACT

Despite intense scholarly interest in entrepreneurial ecosystems, research has not explicitly emphasized ecosystem-level dynamics or explored sources of variance among ecosystems. We theorize that entrepreneurial ecosystems are heterogeneous in a key, system-level characteristic: resilience. We define entrepreneurial ecosystem resilience and theorize that it is intimately tied to the interaction between micro- and macro-level processes of an ecosystem. We contend that entrepreneurial ecosystems differ in both the diversity of participants, ventures, business models, and support organizations, and their coherence around shared values and activities. The interplay between ecosystem diversity and coherence, in turn, produces resilience. Our theory has implications for scholars, entrepreneurs, and policy-makers.

1. Introduction

“The oak fought the wind and was broken, the willow bent when it must and survived.”

— Robert Jordan, *The Fires of Heaven*

Entrepreneurial ecosystems – communities of agents, social structures, institutions, and cultural values that produce entrepreneurial activity – are receiving growing academic, practitioner, and policy-maker attention and are increasingly viewed as a source of economic growth and revitalization (Arruda et al., 2013; Audretsch et al., 2006; Autio and Thomas, 2013; Breznitz and Taylor, 2014; Neck et al., 2004; Spigel, 2016; Spilling, 1996; Van de Ven, 1993). The accumulation of studies is producing a burgeoning literature identifying the key components of entrepreneurial ecosystems (e.g., venture capital, support organizations; Bahrami and Evans, 1995; Kenney and Von Burg, 1999; Isenberg, 2010) and, more recently, teasing apart the connections among system components (Motoyama and Knowlton, 2017; Spigel, 2017). These studies provide important insights into the building blocks of an entrepreneurial ecosystem (EE); however, most research “overlook[s] the heterogeneous nature of ecosystems” and does not acknowledge that they are “highly variegated” (Brown and Mason, 2017: 11); thus, it is still not clear why some EEs thrive, particularly in response to environmental changes, while others decline or fail to develop. Responding to recent calls for a deeper focus on heterogeneity among EEs (Acs et al., 2017), we theorize about a fundamental aspect of EE variation – *resilience*, which we define as *the degree to which an EE can continuously recover from and adapt to exogenous shocks and endogenous pressures* (Cadenasso et al., 2006). Resilience is both a theoretically and practically interesting property because, as we propose, it is an EE-level characteristic that can determine if an ecosystem is able to respond to disruptions, but depends on an EE balancing the seemingly paradoxical tension between the diversity and coherence of its components. Exploring the interplay between resilience, diversity, and coherence, draws

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attention to previously unexamined ecosystem-level properties of EEs, which creates connections to entrepreneurship, organization theory, and economic geography.

The remainder of the paper proceeds as follows. We first discuss the concept of resilience. Since the intellectual roots of resilience are in ecology and the physical sciences, we conclude this overview with a brief discussion of the appropriateness and dangers of applying theory and findings about resilience from biological ecosystems to EEs. We then develop propositions about two sub-characteristics of EEs: diversity and coherence, which we use to motivate our arguments regarding the heterogeneity of EEs in their resilience. Finally, we conclude with a discussion of the implications of our theorizing for EE scholarship and practice.

2. Resilience in ecosystems and entrepreneurial ecosystems

2.1. What is ecosystem resilience?

Studies of ecosystems find that they differ in a property fundamental to their “health” and that reflects the ability to adapt and change in response to disturbances: resilience (Meerow and Newell, 2015). Ecosystems face disturbances arising from shifting external and internal conditions (Holling and Gunderson, 2002; Cadenasso et al., 2006); however, as ecosystem resilience increases, the magnitude of disturbances from which an ecosystem can recover increases and the time it takes for the strained system to adapt decreases (Limburg et al., 2002). An ecosystem that is resilient is able to “absorb” disturbances and undergo the changes necessary to transform its essential behaviors, structures, and identity into a system that is better able to respond to disruptions (Walker et al., 2004).¹ Thus, a resilient ecosystem can survive disturbances without suffering a significant loss of function and, indeed, can be stronger after a disruption (Holling, 1986).

2.1.1. The applicability of the ecosystem's metaphor to EE resilience

Scholars argue that research on EEs should take “the metaphor of the ‘ecosystem’ seriously” (Auerwald, 2015: 1); however, concerns have also surfaced about the applicability of the biological ecosystem metaphor to entrepreneurial ecosystems (e.g., Isenberg, 2016). These concerns are reflective of a long tradition of debates across social science disciplines about the usefulness and danger of using natural metaphors (Hodgson, 2005).

An early application of the biological ecosystems metaphor to business was made by Moore (1993) who argued that “business ecosystems condense out of the original swirl of capital, customer interest, and talent generated by a new innovation, just as successful species spring from the natural resources of sunlight, water, and soil nutrients” (p. 76; cf. Brown and Mason, 2017). Despite the intuitiveness of this assertion, there are differences between physical and entrepreneurial ecosystems, which have implications for conceptualizing EE resilience.

Biological ecosystems (e.g., forests) most notably diverge from socio-economic ecosystems, like EEs, in that the “agents” in biological systems do not have aspirations regarding the functioning of the system and do not formulate plans to meet them. Thus, a challenge in applying biological analogies directly to EEs is that they do not account for individual agency and that EEs are generated by the experimentation and willful actions of entrepreneurs and other ecosystem participants (Read, 2016). In addition, as we argue in the next section, diversity is a necessary component of EE resilience. The variation that breeds diversity in biological ecosystems is often assumed to be random (or quasi-random); in contrast, EEs are human-constructed systems and their diversity is a result of human intentions, choices, and behaviors (Stam and Spigel, 2017).

Differences in the two types of ecosystems have important implications for EE resilience. First, the prevalence of human intention and agency in EEs suggests that social science theories are better-suited for explaining EE resilience than theories imported from the physical sciences. Second, unlike the non-human components of a biological ecosystem, which are not cognizant of the ecosystem as a distinct conceptual entity, the agents of EEs can be aware of the existence of an EE, as well as its diversity, coherence, and resilience. EE participants can actively take steps to change ecosystem-level characteristics. Thus, while there is a deep stream of research on biological resilience (e.g., Walker et al., 2004), its direct application to EEs is approached with caution and the biological interpretation of ecosystems is not taken literally (Stam, 2015).

Although we should use caution in analogizing between biological and entrepreneurial ecosystems, EE behavior is real, not metaphorical, and thus broad similarities can be identified between the two types of ecosystems. For instance, both biological and entrepreneurial ecosystems are systems of interconnected but distinct agents that exhibit self-sustaining properties and behaviors, such as nonlinear dynamics (Roundy et al., 2016). As we argue, the two types of ecosystems also share two general properties, *diversity* and *coherence*, the interplay of which co-produces EE resilience.

2.1.2. The diversity of entrepreneurial ecosystems

Research in economic geography finds that there are benefits to organizations and regional economies from related variety – that is, firms in similar industries locating in the same geographic area (e.g., agglomeration effects; Boschma and Iammarino, 2009; Frenken et al., 2007). There are also distinct benefits from unrelated variety, the co-location of firms from dissimilar industries. For example, “Jacobs’ externalities,” exist in a localized geographic area when a diverse industry mix improves opportunities to interact,

¹ As the paper's opening quotation suggests, there is an important distinction between resilience and resistance. Resistance represents how unyielding an ecosystem is to disturbances. A highly resistant ecosystem will be unresponsive to disturbances. In contrast, a resilient ecosystem is able to *change and respond* to disturbances. Because resilient ecosystems absorb feedback from their environments, resilience is considered a more adaptive characteristic than resistance (cf. Folke, 2006).

copy, modify, and recombine ideas, practices, and technologies across industries (Frenken et al., 2007).

Beyond the spillover benefits of firm diversity, work in portfolio theory (Markowitz, 1991; Morris et al., 2015) suggests that another advantage of a region having a diverse set of firms is that “high sector variety of a regional economy implies that a negative shock in demand for any of these sectors will only have mild negative effects on growth and employment” (Frenken et al., 2007: 686). Thus, diversity in a region's portfolio of firms can be a risk-spreading strategy that guards against unanticipated disruptions.

Research on regional diversity has focused primarily on industry variety. Entrepreneurial ecosystem diversity, however, is broader than industry-differences and is the degree to which an EE contains a broad representation of participant variety (e.g., demographic diversity of EE participants across categories (investors, customers), not just entrepreneurs), venture types (e.g., life-style, aggressive growth ventures; cf. Morris et al., 2015), business models, and support organizations (e.g., for-profit and nonprofit-oriented incubators and accelerators).

Diversity can manifest in the types of ventures in an EE; however, its genesis is at the individual – “micro” – level of the EE. In addition to variation in individual characteristics (e.g., business experiences, demographics, intentions), participants can also vary across EEs in the degree to which they emphasize experimentation in the venture creation process (Baum et al., 2011). A growing movement among entrepreneurship practitioners promotes trial-and-error approaches (methods often grouped under folk practices such as “Lean Startup” (Blank, 2013), “Design Thinking,” (Martin, 2009), and “Agile” (Martin, 2002)). The goal of these approaches is to increase entrepreneurial discovery and alignment with market forces. For instance, entrepreneurs utilize experimentation approaches in the search for their business models and in seeking a close fit between the products they are developing and consumer demand. A consequence of experimentation, and its unpredictability, is that it can produce novelties in business models, products, and types of ventures in the ecosystem (Metcalf, 2005), which increase overall ecosystem diversity. An EE with a wide array of support organizations, spanning the nonprofit to for-profit continuum, can also influence the diversity of ventures created (Roundy, forthcoming). In sum, the variation across EEs in both the characteristics of participants, the extent to which participants engage in experimentation-based entrepreneurial processes, and the focus of support organizations, suggests:

Proposition 1a. Entrepreneurial ecosystems are heterogeneous in the diversity of their participants, types of ventures, business models, and support organizations.

2.1.3. The coherence of entrepreneurial ecosystems

While diversity represents an important source of heterogeneity across EEs, the forces that promote *coherence* are also critical to ecosystem functioning. Research examining entrepreneurial ecosystems has hinted at the importance of EE coherence, but direct theorizing about its role has not occurred. For example, Stam (2015, p. 1765) offers a broad definition of EEs as a “set of interdependent actors and factors *coordinated* in such a way that they enable productive entrepreneurship” (emphasis added). Similarly, Mason and Brown (2014, p. 5) propose a definition of EEs as “a set of interconnected entrepreneurial actors, entrepreneurial organizations, institutions and entrepreneurial processes which formally and informally *coalesce* to connect, mediate and govern the performance within the local entrepreneurial environment” (emphasis added). Motoyama and Knowlton (2017: 20) emphasize the “connectivity” between EE elements. Findings suggest that coherence matters for EE functioning. For instance, in a study of the Phoenix, Arizona EE, Mack and Mayer note the fragmentation among support organizations (an example of a lack of coherence) as an impediment to EE development and that policy-makers desire to increase cohesion. But prior research has not explicitly examined what coherence represents or why it is important in the context of EEs.

Coherence is the degree of association between the components of an EE, which causes them to coalesce into an interconnected group (i.e., the ecosystem). Coherence creates the structure that gives form to an EE; thus, it exists at the “macro-level” of the system. Indeed, for a group of individual agents and organizations to be an EE, rather than simply a group of unconnected, autonomous components, entrepreneurs and other participants must operate according to some degree of common intentions, behaviors, and values.

EE coherence has several potential sources. Entrepreneurs and other ecosystem participants can share broad intentions and goals, such as founding and scaling new ventures. These similarities will cause them to engage in common behaviors and activities (e.g., seeking investors and customers; testing their business models) and to pursue common outcomes (e.g., becoming cash flow positive). Coherence can also arise from the novelties entrepreneurs introduce into the EE (Muñoz and Encinar, 2014). For example, a business model innovation that proves successful can be copied and used by other entrepreneurs in an EE, which produces similarities across ventures. Lastly, coherence can be generated by an EE's agents sharing common values, logics, and cultural rule sets, such as “give [resources to the ecosystem] before taking,” “help other ecosystem participants,” and “be tolerant of entrepreneurial failure” (Axelrod, 1997; Feld, 2012). Such coherence can be produced by the general effects of the “institutional thickness” of an EE (cf. Amin and Thrift, 1995) and by more specific dynamics involving an EE's support organizations.

Support organizations (e.g., incubators) increase the coherence of EE participants' intentions, behaviors, and outcomes by exposing participants to common logics: “the formal and informal rules of action, interaction and interpretation that guide and constrain decision makers” (Ocasio and Thornton, 1999: 804). Two logics are theorized to be dominant in functioning entrepreneurial ecosystems (Roundy, forthcoming): an entrepreneurial-market logic, which is a related set of goals and behaviors focused on innovation, the creation of new markets, business models, and technologies, the pursuit of opportunities in spite of resource scarcity, and tolerance for uncertainty and failure (Cunningham et al., 2002), and a community logic, which is a focus on community needs, development, prosperity, trust, and cooperation (Marquis et al., 2011; Thornton et al., 2012). The community logic is especially important for developing trust between EE participants that enables knowledge sharing and learning and decreases transaction costs among EE members. Trust also manifests as a form of “suspension of disbelief” tacitly subscribed to by participants in the EE when

helping early-stage entrepreneurs. For example, if there is coherence in an EE, when entrepreneurs within the ecosystem meet, there will be a degree of trust simply because of their association with the EE.

Through support organizations, which must often internally balance entrepreneurial market and community logics (Roundy, forthcoming), and other mechanisms by which EE participants are exposed to common logics (e.g., informal events), logics spread and become the shared values, conventions, routines, and simple rules that guide ecosystem participants, give form to the entrepreneurial community, and create the social and cultural fabric that causes the EE to function like an interdependent system rather than a loose collection of agents.

In sum, any force that strengthens the association between an EE's components will increase its coherence, act as a force that helps to order it and, as we theorize in the next section, ward against dissipative or disruptive tendencies. However, in the same way that EEs differ in their degree of diversity, they also differ in the strength of their cohering forces.

Proposition 1b. Entrepreneurial ecosystems are heterogeneous in their degree of coherence.

2.2. Diversity, coherence, and entrepreneurial ecosystem resilience

An important finding from studies of biological and social ecosystems is that there is a strong, positive relationship between the diversity of an ecosystem's elements and its resilience (e.g., [Manson, 2001](#)). Counterbalancing and contributing to the influence of diversity, coherence in the components of an ecosystem can generate sufficient structure for an EE to “bend without breaking” and to survive and adapt to disruptions. Thus, there is both a tension and a synergistic effect between the dissimilarity-producing forces of diversity and an EE's cohering forces.

The tension in EEs between the diversity and coherence of components is akin to the tension that organizations, as open systems, must balance between the differentiation and integration of their sub-units (e.g., [Lawrence and Lorsch, 1967](#)). Both entities must incorporate unique actions that effectively respond to changing market and environmental conditions. These differentiating actions must be assimilated into the system as a whole; however, whereas organizations require integration across functional areas and individual departments, EEs necessitate a melding of new activities among the various EE components (e.g. incubators, funding sources, etc.). In diverse EEs, there is a greater likelihood for ecosystem adaptation to shifting environmental demands, as there are more novelty and variety on which market forces can find alignment. However, as a type of integration mechanism, EEs require coherence to pass along key ecosystem properties through cohering sets of intentions and rules.

The diversity of an EE's participants on various dimensions (e.g., demographic characteristics; entrepreneurial intentions) is associated with the ecosystem's agents, collectively, having a greater breadth of knowledge (e.g., [Folke et al., 2005](#)). A robust knowledge base, with rich and varied schema, can increase the number of responses available to the EE. In contrast, coherence provides the shared values and patterns of thinking required for the underlying structure of the system to survive and support such adaptations. Furthermore, this line of argumentation suggests that shared values and sources of coherence, such as tolerance for entrepreneurial failure and the collective desire to help individuals navigate the entrepreneurial process, will increase the experimentation that occurs in the EE and, in turn, increase ecosystem diversity. Thus, coherence and diversity, reinforce one another. In sum, coherence represents a complex latticework of connections among the diverse components of an EE that allows the ecosystem to absorb a disturbance and adapt while passing along key ecosystem properties.

Proposition 2. Diversity and coherence will co-produce entrepreneurial ecosystem resilience.

The proposed differences among EEs in diversity and coherence, and variation in EEs' abilities to manage the balance between diversity and coherence, can be expected to produce variation in EE resilience.

Proposition 3. Inter-ecosystem differences in diversity and coherence will produce heterogeneity in entrepreneurial ecosystem resilience.

3. Discussion and conclusion

3.1. Implications for entrepreneurial ecosystems scholarship and practice

The literature on EEs, while thriving, has not examined ecosystem-level properties. By examining EEs *as ecosystems*, our theorizing shifts the focus from the identification of individual ecosystem components to a view of the EE as a system that must respond to disruptions by changing, adapting, and evolving over time. In doing so, we add to other recent work emphasizing the evolutionary development of EEs rather than static approaches (e.g., [Mack and Mayer, 2016](#)). This holistic view, and our theorizing about EE resilience, has several implications for scholars and policy makers attempting to stimulate the EE development.

First, this paper draws attention to two critical forces in EEs – diversity and coherence – which create a necessary source of tension. Furthermore, this tension may lead to an interesting paradox: certain shared values at the system level, such as tolerance for error, encourage entrepreneurial experimentation leading to diversity in business models and types of ventures. But as the diversity of an EE increases, this dissimilarity can make it more difficult for the ecosystem to cohere. At what point does this tension become unbalanced? Our theorizing suggests that an important facet of understanding EEs is determining how tightly coupled the forces of diversity and coherence are at different levels and scales of the system and how changes in the degree of tension in these forces can influence EE resilience and behaviors.

Second, our theorizing has implications for attempts to measure the functioning of EEs. Work on ecosystems in other contexts has found that, because of their complexity, they cannot be effectively assessed using simple, “count-based” metrics. Levin (1998) describes this challenge in the context of biological ecosystems and explains that, for example, when attempting to measure an ecosystem's diversity, “simple species counts [...] do not alone capture the features that are most important for sustaining ecosystem functioning [as] [n]ot all species are equally important to the maintenance of key processes” (429). Although we acknowledge the important differences between biological and entrepreneurial ecosystems, we would argue that this statement is at least partially applicable to EEs. Traditional metrics used to measure entrepreneurial activity, such as the number of jobs created or counts of firms founded, may provide a baseline, but do not fully capture the health of the ecosystem.

Likewise, if the resilience of an EE is an important indicator of its functioning, and if resilience depends on factors such as experimentation, diversity, and coherence, then policy-makers and support organizations should invest in programs and initiatives that promote these qualities among EE participants. Moreover, if participants want to ensure that an EE can respond and adapt to disturbances, they should recognize that it may be dangerous to create programs or policies that limit experimentation or over-invest in one specific type of entrepreneurship, technology, or industry. Policy-makers might similarly benefit from considering how an ecosystem maintains and increases the coherence among agents.

These arguments suggest that policy-makers and other EE caretakers would be wise to measure EE characteristics that represent the ecosystem's diversity and coherence. For instance, as a starting point, Stangler and Bell-Masterson (2015) have proposed assessing the diversity in an EE by measuring the number of economic specializations, income mobility of participants, and percentage of immigrants, and propose potential sources of diversity data (e.g., the Quarterly Census of Employment and Wages). The coherence of an EE is perhaps more difficult to assess. Stangler and Bell-Masterson suggest measuring EE connectivity (i.e., connections between elements in an EE). However, coherence is more related to the *strength* and *type* of connections than the absolute number. Thus, to measure coherence could involve surveying ecosystem participants about their shared beliefs, degree of collaboration, and willingness to trust other EE participants. Developing such measures would be the first step towards assessing EE resilience.

3.2. Conclusion

Every region has some level of entrepreneurial activity and a growing number have entrepreneurial ecosystems. However, not every ecosystem is resilient. The literature on biological ecosystems provides a starting point for a holistic view of EEs that extends the component-level view of past research and acknowledges ecosystem differences. Building on this work, our aim was to draw attention to EE resilience, an understudied, but vital, ecosystem-level property of EEs. We hope doing so stimulates new avenues for studying EEs, a phenomenon increasingly viewed as an engine for economic and community development.

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